

Table 1: How open source tools enhance data management and security

Cost-effective data management	Open source tools provide cost advantage by being freely available, unlike proprietary software with high licensing fees. They allow organisations to allocate their budgets more efficiently and provide an opportunity for smaller companies and startups to access powerful data management solutions without breaking the bank.
Customisability	Open source tools offer a high degree of customisability. Organisations can tailor the software to their specific needs, creating solutions that fit unique data requirements. With open source tools, you're not bound by the limitations of proprietary software, and you can adapt the tools to your evolving data needs.
Collaboration and community support	Open source tools benefit from extensive user communities that contribute to their development and support. When you use open source tools, you have access to a community of users who can provide advice, solutions to issues, and share best practices for data management and security.
Security through transparency	Open source tools are considered more secure due to their transparency. The source code is open for review, making it difficult for malicious actors to insert hidden vulnerabilities or backdoors. Security experts worldwide scrutinise the code, and any vulnerabilities or weaknesses are quickly identified and patched. This open, transparent development process contributes to the overall security of open source solutions.
Rapid updates and fixes	Open source tools typically have a shorter response time to security vulnerabilities. When a flaw or bug is discovered, the open source community can respond quickly by releasing patches or updates to address the issue. This rapid response is crucial in an era where cyber threats are constantly evolving, and data breaches can have severe consequences.
Integration with other tools	Open source tools are designed to be highly integrative, facilitating their use in conjunction with other software and systems. Their ability to connect and collaborate with various tools is particularly important in data management, as it enables organisations to build comprehensive data ecosystems that cater to their unique requirements.

Apache Hadoop

- **Data management:** Hadoop is a powerful framework for storing and processing large data sets. It provides distributed storage and processing capabilities that help organisations effectively manage their Big Data, ensuring data integrity and accessibility.
- **Security:** Hadoop offers several security features, such as authentication, authorisation, and encryption, to protect sensitive data stored within the platform.

Elasticsearch

- **Data management:** Elasticsearch is a distributed, RESTful search and analytics engine that excels at indexing and searching large volumes of data. It's widely used for log and event data management, making it easier to uncover valuable insights from various data sources.
- **Security:** Elasticsearch has robust security features, including role-based access control and data encryption, to ensure that sensitive information is safe from unauthorised access.

OpenStack

- **Data management:** OpenStack is an open source cloud computing platform that enables organisations to build and manage private and public clouds. It offers storage solutions like Cinder (block storage) and Swift (object storage) for efficient data management.
- **Security:** OpenStack includes security features like identity and access management, firewall as a service, and encryption, providing a secure infrastructure for data storage and processing.